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Assembly system and method for packaging web material in a roll

Abstract

Rolling assembly for packaging web material in a roll having a base member with concave arcuate feed surface defining an upwardly-extending ramp, first intermediate member movably coupled to base member, and top member moveably coupled to at least one of base member or first intermediate member, top member having a concave arcuate pressure surface facing feed surface of base member. The base member, first intermediate member and top member collectively forming an iris with a feed space, the iris moveable between a first condition and a second condition, and configured to receive web material along the feed surface and direct the web material upwardly toward the top member to form a roll within the feed space, the iris moving toward a second condition as the roll of web material increases in cross dimension against the pressure surface of the top member.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 16/841,386 filed Apr. 6, 2020, which claims priority to U.S. Provisional Application Ser. No. 62/829,488 filed Apr. 4, 2019, the contents of each of which is hereby incorporated by reference in its entirety.

BACKGROUND

Field of the Disclosed Subject Matter

(1) The presently disclosed subject matter relates generally to a rolling assembly, and related system and method of using the same, for packaging web material in a roll for transport, storage, and commercialization.

DESCRIPTION OF RELATED ART

(2) A wide variety of web materials are commercialized for various uses. For example, bags and resealable packages for containing materials such as food, household items, and waste are often formed from webs of plastic material. Such packages are inexpensive, lightweight, and easily manufactured in large quantities.

(3) Often web materials, or the products produced therefrom, need to be transported, stored, or commercialized. One known solution is to create a roll of the web material. For example, the web material can be rolled around a shaft or spindle. However, transporting or packaging the web material with a shaft or spindle is not always desirable. For example, the shaft or spindle may add material costs, create additional waste, and increase the weight of the finished roll of web material.

(4) Web material can be rolled without the use of a shaft or spindle; however, known methods can result in a roll that is loosely rolled, or which has a larger diameter for a given amount of web material than desired. Therefore, there continues to be a need for improved assemblies, systems, and methods for packaging web materials in a tight roll for efficient transport, packaging, and commercialization.

SUMMARY

(5) The purpose and advantages of the disclosed subject matter will be set forth in and apparent from the description that follows, as well as will be learned by practice of the disclosed subject matter. Additional advantages of the disclosed subject matter will be realized and attained by the assemblies, methods, and systems particularly pointed out in the written description and claims hereof, as well as from the appended drawings.

(6) To achieve these and other advantages, and in accordance with the purpose of the disclosed subject matter, as embodied and broadly described, the disclosed subject matter includes a rolling assembly for packaging web material in a roll. The rolling assembly has a base member with a concave arcuate feed surface defining an upwardly-extending ramp and a first intermediate member moveably coupled to the base member. The assembly further includes a top member moveably coupled to at least one of the base member or the first intermediate member. The top member has a concave arcuate pressure surface facing the feed surface of the base member. The base member, the first intermediate member and the top member form an iris with a feed space defined between the feed surface and the pressure surface. The iris is moveable between a first condition and a second condition, and the feed space has a first cross dimension in side view between the feed surface and the pressure surface in the first condition. The feed space has a second cross dimension in side view between the feed surface and the pressure surface when the iris is in the second condition. The second cross dimension is greater than the first cross dimension. The iris is configured in the first condition to receive web material along the feed surface and direct the web material upwardly toward the top member to form a roll within the feed space. The iris is configured to move toward the second condition as the roll of web material increases in cross dimension against the pressure surface of the top member.

(7) The disclosed subject matter also includes a system for packaging web material in a roll. The system includes a feed assembly configured to carry a stream of web material, and a rolling assembly proximate the feed assembly and configured to receive a stream of web material. In accordance with the disclosed subject matter, the rolling assembly includes the features described above.

(8) The disclosed subject matter also includes a method of packaging web material in a roll. Methods in accordance with the disclosed subject matter include providing a rolling assembly configured to receive a stream of web material. In accordance with the disclosed subject matter, the rolling assembly includes the features described above. Methods in accordance with the disclosed subject matter further include delivering a stream of web material to the assembly. The iris of the rolling assembly receives the web material along the feed surface and directs the web material upwardly toward the top member to form a roll within the feed space. The iris moves toward the second condition as the roll of web material increases in cross dimension against the pressure surface of the top member.

(9) As recognized in the art, the assemblies, systems, and methods disclosed herein can include some or all of the features described herein, or any suitable combination thereof. It is to be understood that both the foregoing general description and the following detailed description are exemplary and are intended to provide further explanation of the disclosed subject matter claimed.

(10) The accompanying drawings, which are incorporated in and constitute part of this specification, are included to illustrate and provide a further understanding of the containers and methods of the disclosed subject matter. Together with the description, the drawings serve to explain the principles of the disclosed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of an exemplary embodiment of a rolling assembly for packaging web material in accordance with the disclosed subject matter.
- (2) FIG. 2 is a side view of the rolling assembly of FIG. 1 with the iris in a first condition.
- (3) FIG. 3 is a side view of the rolling assembly of FIG. 1 with the iris in a second condition.
- (4) FIG. 4A is a side view of the base member of the rolling assembly of FIG. 1.
- (5) FIG. 4B is a front view of the base member of the rolling assembly of FIG. 1.
- (6) FIG. 5 is a side view of the first intermediate member of the rolling assembly of FIG. 1.
- (7) FIG. 6 is a side view of the top member of the rolling assembly of FIG. 1.
- (8) FIG. 7 is a side view of the second intermediate member of the rolling assembly of FIG. 1.
- (9) FIG. 8A is a side view of the rolling assembly of FIG. 1 with the iris in a first condition to receive a stream of web material in a shingled arrangement.
- (10) FIG. 8B is a side view of the rolling assembly of FIG. 8A with the iris in a first condition forming a roll of web material within the feed space.
- (11) FIG. 8C is a side view of the rolling assembly of FIG. 8A with the iris moving toward a second condition as the roll of web material increases in cross dimension.
- (12) FIG. 9A is a side view of another exemplary embodiment of a rolling assembly for packaging web material in accordance with the disclosed subject matter with the iris in a first condition.
- (13) FIG. 9B is a side view of the rolling assembly of FIG. 9A with the iris in a second condition.
- (14) FIG. 10A is a side view of another exemplary embodiment of a rolling assembly for packaging web material in accordance with the disclosed subject matter with the iris in a first condition.
- (15) FIG. 10B is a side view of the rolling assembly of FIG. 10A with the iris in a second condition.
- (16) FIG. 11A is a side view of another exemplary embodiment of a rolling assembly for packaging web material in accordance with the disclosed subject matter with the iris in a first condition.
- (17) FIG. 11B is a side view of the rolling assembly of FIG. 11A with the iris in a second condition.
- (18) FIG. 12A is a side view of another exemplary embodiment of a rolling assembly for packaging web material in accordance with the disclosed subject matter with the iris in a first condition.
- (19) FIG. 12B is a side view of the rolling assembly of FIG. 12A with the iris in a second condition.
- (20) FIG. 13 is a perspective view of the rolling assembly of FIG. 12A and a feed assembly for delivering web material to the rolling assembly.
- (21) FIG. 14 is a perspective view of two exemplary rolling assemblies in accordance with the disclosed subject matter arranged in a parallel configuration and a feed assembly for delivering web material to the rolling assemblies.

DETAILED DESCRIPTION

- (22) Reference will now be made in detail to the various exemplary embodiments of the disclosed subject matter, exemplary embodiments of which are illustrated in the accompanying drawings. The structure and corresponding method of operation of the disclosed subject matter will be described in conjunction with the detailed description of the rolling assembly and system.
- (23) The rolling assembly, system, and method presented herein can be used for the packaging transport, storage, and commercialization of a wide variety of web material. The disclosed subject matter is particularly suited for efficiently packaging web formed food storage containers, such as slider bags, in a roll.
- (24) In accordance with the disclosed subject matter herein, the rolling assembly generally includes a base member having a concave arcuate feed surface defining an upwardly-extending ramp and a first intermediate member moveably coupled to the base member. The assembly further includes a top member moveably coupled to at least one of the base member or the first intermediate member. The top member has a concave arcuate pressure surface facing the feed surface of the base member.

The base member, the first intermediate member and the top member collectively form an iris with a feed space defined between the feed surface and the pressure surface. The iris is moveable between a first condition and a second condition, and the feed space has a first cross dimension in side view between the feed surface and the pressure surface in the first condition and a second cross dimension in side view between the feed surface and the pressure surface in the second condition. The second cross dimension is greater than the first cross dimension. The iris is configured in the first condition to receive web material along the feed surface and direct the web material upwardly toward the top member to form a roll within the feed space. The iris configured to move toward the second condition as the roll of web material increases in cross dimension against the pressure surface of the top member.

(25) The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views, serve to further illustrate various embodiments and to explain various principles and advantages in accordance with the disclosed subject matter. For purpose of explanation and illustration, and not limitation, exemplary embodiments of the container in accordance with the disclosed subject matter are shown in FIG. 1-FIG. 14. The assembly of the disclosed subject matter is suitable for use with a wide variety of web formed articles. As used herein, the terms “front,” “rear,” “side,” “top,” and “bottom” are used for the purpose of illustration only, and not limitation. That is, it is recognized that the terms “front,” “rear,” “side,” “top,” and “bottom” are merely used herein as a point of reference, and can alter based upon perspective.

(26) For purpose of illustration, and not limitation, reference is made to the exemplary embodiment of an assembly **10** shown in FIG. 1-FIG. 8C. In accordance with the disclosed subject matter, assembly **10** includes a base member **100**, a first intermediate member **200** moveably coupled to the base member **100**, and a top member **300** moveably coupled to at least one of the base member **100** or the first intermediate member **200**. As described further herein, the base member **100** includes a concave arcuate feed surface **105** defining an upwardly-extending ramp, and the top member **300** includes a concave arcuate pressure surface **305** facing the arcuate feed surface **105**. In accordance with the disclosed subject matter, and as described further herein, the base member **100**, first intermediate member **200**, and the top member **300** collectively form an iris **500** with a feed space **510** defined between the feed surface **105** and the pressure surface **305**.

(27) With reference to FIG. 4A, the base member **100** can have a first end **101** and a second end **102**. A feed surface **105** can extend along at least a portion of the base member **100** between the first end **101** and the second end **102** to define an upwardly-extending ramp. As embodied herein, the feed surface **105** can have a radius of curvature and can define a crescent-like shape in side view. The radius of curvature can be a constant radius of curvature, or the radius of curvature can be complex, such that the radius of curvature varies between the first end **101** and the second end **102**. As described further herein, the radius of curvature of the feed surface **105** can be selected according to the properties of the web formed articles to be packaged.

(28) Additionally or alternatively, and as embodied herein, the feed surface **105** of the base member **100** can include an angled or sloped region **110**, which can extend along at least a portion of the feed surface **105** between the first end **101** and the second end **102**. The angled or sloped region **110** can be disposed at an angle α in side view relative to a bottom edge **104** of the base member **100**, as shown in FIG. 4A. Additionally, and as further embodied herein, the base member **100** can have a width dimension **103** in end view, such as depicted in the front view of FIG. 4B.

Furthermore, and as embodied herein, the feed surface **105** can include chamfered edges **111** along one or more sections of the feed surface, such as between an outer face of tier **106** towards a center region **109**.

(29) Further referencing FIG. 4B, the base member **100** can include one or more tiers different from each other in end view, each of which can facilitate the articulating motion of a respective one of the top member **300** and one or more intermediate members **200**, **400**, as further described

herein. In accordance with one aspect of the disclosed subject matter, and as embodied herein, the base member **100** can include an outer tier **106**, a middle tier **107**, and a lower tier **108**, each defining a respective outer surface portion of the base member **100**. For purpose of example and not limitation, and as embodied herein, outer tier **106** can extend to a portion proximate the first end **101** of the base member **100** and lower tier **108** can be disposed proximate a second end **102** of the base member with middle tier **107** disposed in a region therebetween. The distance or depth between adjacent outer surface portions of the base member **100**, as defined by each respective tier can be selected based on the thickness of the corresponding top member or intermediate members to be coupled thereto, as further described herein. As further embodied herein, the base member **100** can also include connection points **115**, **120**, and **130**, which can be used to couple the top member **300** and one or more intermediate members **200**, **400** to the base member **100**, as further described herein. For purpose of example, and not limitation, and as embodied herein, the connection points can define one or more holes which can receive a fastener, such as a bolt, screw, peg, nail, rivet, or the like. Alternatively, such fasteners can be formed integral with the base member.

(30) With reference to FIG. **1** and FIG. **5**, the assembly **10** further includes a first intermediate member **200** moveably coupled to the base member **100**. For example, and as embodied herein, the first intermediate member can be pivotally connected to the base member **100** as depicted in the exemplary embodiment of FIG. **1**-FIG. **8C**. For purpose of example, and as embodied herein, the first intermediate member **200** can include a connection point **203** and the first intermediate member **200** can be pivotally connected to the base member **100** at connection point **203**. For purpose of example, and not limitation, connection point **203** can define one or more holes configured to receive a fastener, such as a bolt, screw, peg, nail, or rivet to pivotally connect the first intermediate member **200** to the bottom member **100**. The intermediate member can further include a first end **201** and a second end **202**.

(31) In accordance with another aspect of the disclosed subject matter, and as further embodied herein, the first intermediate member **200** can include a connection point **204** such as depicted in the exemplary embodiment of FIG. **1**-FIG. **8C**. Connection point **204** can be configured, for example, to slidably connect the first intermediate member **200** with a second intermediate member **400**, as further described herein. Additionally, or alternatively, and in accordance with another aspect of the disclosed subject matter, the first intermediate member **200** can also include a slot **206** to facilitate movement of the first intermediate member **200** relative to the top member **300**. For purpose of example, and as embodied herein, slot **206** can be configured to slidably connect the first intermediate member **200** with the top member **300**.

(32) In accordance with another aspect of the disclosed subject matter, and as described further herein, the intermediate member **200** can include an arcuate first intermediate pressure surface **205** extending along at least a portion of the intermediate member **200** between the first end **201** and the second end **202**. As described previously with respect to the feed surface **105**, the first intermediate pressure surface **205** can have a radius of curvature to define a crescent-like shape in side view. The radius of curvature can be a constant radius of curvature, or complex. As embodied herein, and as described further below, the first intermediate member **200** collectively with the base member **100** and top member **300** can define an iris **500** to apply pressure on the roll of web material **25** as the roll increases in cross dimension and the iris **500** expands from a first condition to a second condition.

(33) With reference to FIG. **1** and FIG. **6**, the rolling assembly **10** further includes a top member **300** moveably coupled to at least one of the base member **100** and the first intermediate member **200**. In accordance with the disclosed subject matter, the top member **300** includes a pressure surface **305**. As embodied herein, the top member can include a first end **301** and a second end **302**, and the pressure surface **305** can extend along at least a portion of the top member between the first end **301** and the second end **302**. In accordance with the disclosed subject matter, and with

reference to FIG. 1, the pressure surface **305** of the top member **300** faces the feed surface **105** of the base member **100**. As embodied herein, the pressure surface **305** can have a radius of curvature to define a crescent-like shape in side view. As previously noted with respect to the base member **100**, the radius of curvature of the arcuate pressure surface **305** can be constant or complex. For purpose of example and not limitation, the radius of curvature of the top member **300** can be approximately 2" to approximately 5". As further embodied herein, the pressure surface **305** can have a width dimension, which can include chamfered edge portions along some or all of the length if desired. For purpose of example and not limitation, the pressure surface **305** can have a width dimension of between approximately 0.5" and approximately 0.75".

(34) In accordance with the disclosed subject matter, the top member **300** can be moveably coupled to at least one of the base member **100** or the first intermediate member **200**. For example, the top member can be pivotally connected to the base member **100**. For purpose of example, and as embodied herein, the top member **300** can include connection point **303**, which can define one or more holes configured to receive a fastener, such as a bolt screw, peg, nail, or rivet, to pivotally connect the top member **300** to the base member **100**. In accordance with another aspect of the disclosed subject matter, and as described further herein, the top member **300** can further include a second connection point **304**, as described further below.

(35) In accordance with the disclosed subject matter, the base member **100**, first intermediate member **200**, and the top member **300** collectively form an iris **500** with a feed space **510** defined between the feed surface **105** and the pressure surface **305**. The iris **500** is depicted in broken line in FIGS. 2 and 3 for purpose of illustration only. As described further herein, the iris **500** is moveable between a first condition and a second condition. Referencing FIG. 8A, the iris **500** of the exemplary embodiment of FIG. 1-FIG. 8C is depicted in a first condition. In accordance with the disclosed subject matter, the feed space **510** has a first cross dimension in side view measured between the arcuate feed surface **105** and the arcuate pressure surface **305** when the iris **500** is in a first condition.

(36) Further referencing FIG. 8A, the iris **500** is configured in the first condition to receive web material **20** along the arcuate feed surface **105**. In accordance with the disclosed subject matter, the iris **500** is configured to direct the web material **20** from the first end **101** upwardly toward the top member **300** to form a roll **25** within the feed space **510**. For purpose of example, and not limitation, directional arrow **22** indicates the direction of movement of the web material **20** within the feed space **510**. With reference to FIG. 8A and FIG. 8B, the leading edge **21** of the web material **20** is first directed upwardly by the shape of the iris **500** in the first condition, and then wrapped back to fold over onto the top of the adjacent web material **20** to initiate a roll **25**.

(37) In accordance with the disclosed subject matter, the roll **25** can increase in cross dimension as the web material **20** continues to enter the feed space **510** in the direction indicated by arrow **35**. The iris **500** moves towards a second condition as the roll of web material **25** increases in cross dimension against the pressure surface **305** of the top member **300**. With reference to FIG. 3, the iris **500** of the exemplary embodiment of FIG. 1-FIG. 8C is depicted in a second condition, and in accordance with the disclosed subject matter, a second cross dimension can be measured in side view between the feed surface **105** and the pressure surface **305** when the iris **500** is in a second condition. The second cross dimension is greater than the first cross dimension of the feed space measured when the iris **500** is in a first condition.

(38) Additionally, and in accordance with the disclosed subject matter, the iris **500** can be configured to apply pressure to the roll of web material **25** as the iris moves between a first condition and a second condition. For purpose of example, and not limitation, the amount of pressure the iris **500** applies to the roll **25** can be controlled in a number of ways, such as by selecting the weight of the top member **300**, incorporating a spring bias between the various members, or by adjusting the friction fit between the various members. As discussed further herein, weight of the top member **300** can be controlled by adding one or more ballast weights to the top

member. Additionally or alternatively, the material of the top member **300** can be selected to achieve the desired weight and apply the desired pressure to the roll of web material **25** as the iris moves between the first condition and the second condition. For example, and as embodied herein, the top member **300** can be formed of steel.

(39) Additionally, and as embodied herein, friction fit between members can be adjusted by adjusting the tightness of fasteners connecting the first intermediate member **200** to the base member and connecting the top member **300** to at least one of the first intermediate member **200** or the base member **100**. Adjusting the tightness of the fasteners can control the force required to move the iris **500** from a first condition to a second condition as the roll of web material **25** increases in cross dimension. For purpose of example, and as embodied herein, the bolts pivotally connecting the first intermediate member **200** and the top member **300** to the base member can be tightened to increase the amount of pressure the iris **500** exerts on the roll of web material **25** as the roll increases in cross dimension against the arcuate pressure surface of the top member.

(40) In accordance with another aspect of the disclosed subject matter, and as further embodied in the exemplary embodiment of FIG. 1-FIG. 8C, the assembly **10** can further include a second intermediate member **400** moveably coupled to the base member **100**. As embodied herein, the second intermediate member **400** can further define the movement of the iris **500** and the rolling assembly **10**. With reference to FIG. 7, and as embodied herein, the second intermediate member can include a first end **401** and a second end **402**. An arcuate second intermediate pressure surface **405** can extend along at least a portion of the second intermediate member **400** between the first end **401** and the second end **402**. As described above with respect to the feed surface **105**, the second intermediate pressure surface **405** can have a radius of curvature to define a substantially crescent-like shape in side view. The radius of curvature can be a constant radius of curvature, or complex. As embodied herein, the second intermediate pressure surface **405** can further define the iris **500** and can apply pressure on the roll of web material **25** as the roll increases in cross dimension and the iris **500** expands from a first condition to a second condition.

(41) In accordance with another aspect of the disclosed subject matter, and as further embodied in the exemplary embodiment of FIG. 1-FIG. 8C, the first intermediate member **200**, second intermediate member **400** and top member **300** can each be pivotally connected to the base member. For example, and as embodied herein, the top member **300** can be pivotally connected at lower tier **108**, the first intermediate member **200** can be pivotally connected at middle tier **107**, and the second intermediate member **400** can be pivotally connected at outer tier **106**. In this manner, the top member **300**, the first intermediate member **200**, and the second intermediate member **400** can be stacked relative to each other and free to move and/or pivot relative to the base member without interfering with the other members.

(42) As embodied herein, the top member **300** can be pivotally connected by connection points **115** and **303** on lower tier **108**, the first intermediate member can be connected by connection points **120** and **203** on middle tier **107** of the base member **100**, and the second intermediate member **400** can be pivotally connected by connection points **130** and **403** on outer tier **106**. As described above, the depth of each tier on the base member **100** can correspond to a thickness dimension of the respective intermediate member or top member such that the intermediate and top members can pivot with respect to the base member **100** without interfering with one another.

(43) In accordance with another aspect of the disclosed subject matter, and as further embodied in the exemplary embodiment of FIG. 1-FIG. 8C, the top member **300** and/or the second intermediate member **400** can each be slidably connected to the first intermediate member **200**. For purpose of example, and as embodied herein, the top member **300** can include a connection point **304** which can define a hole configured to receive a fastener, such as a bolt, screw, peg, pin, nail, or rivet, that slidably interfaces with slot **206** in the first intermediate member **200**. Furthermore, the first intermediate member **200** can include a connection point **204** and a slot **206**. Connection point **204** can include a fastener, such as a bolt, screw, peg, pin, nail, or rivet, that slidably interfaces with a

slot **406** in the second intermediate member **400**. The slidable connections between the top member **300**, second intermediate member **400**, and first intermediate member **200** can further define the motion of the iris **500** as the iris **500** moves from a first condition to a second condition.

(44) In accordance with another aspect of the disclosed subject matter, and with reference to the exemplary rolling assembly **1100** depicted in FIGS. **9A** and **9B**, the first intermediate member **1120** can be pivotally connected to the base member **1110**, and the top member **1130** can be pivotally connected to the first intermediate member **1120** to define iris **1150** therebetween. For example, top member **1130** can include a first end **1131** pivotally connected to the intermediate member **1120** and a free end **1132** with a concave arcuate pressure surface **1135** therebetween.

(45) In accordance with the disclosed subject matter, the base member **1110**, the first intermediate member **1120** and the top member **1130** collectively form an iris **1150** with a feed space **1151** defined between the feed surface **1105** and the pressure surface **1135**. The iris **1150** of the exemplary rolling assembly **1100** is moveable between a first condition, as depicted in FIG. **9A**, and a second condition, as depicted in **9B**. With reference to FIG. **9A**, the feed space **1151** has a first cross dimension in side view between the feed surface **1105** and the pressure surface **1135** in the first condition. With reference to FIG. **9B**, the feed space **1151** has a second cross dimension in side view between the feed surface **1105** and the pressure surface **1135** in the second condition, the second cross dimension being greater than the first cross dimension. The iris **1150** is configured to receive web material along the feed surface **1105** of the base member **1110** in the first condition and direct the web material upwardly toward the top member **1130** to form a roll within the feed space **1151**. The iris **1150** is configured to move toward the second condition as the roll of web material increases in cross dimension against the pressure surface **1135** of the top member **1130**.

(46) In accordance with another aspect of the disclosed subject matter, the rolling assembly **1100** can be moveable to an extraction condition. For purpose of example and as embodied herein, the first intermediate member **1120** can pivot about connection point **1115** on base member **1110** to extract a finished roll of web material from the rolling assembly **1100**. The free end **1132** of the top member **1130** can be configured to urge the roll of web material out of the feed space **1151** when the rolling assembly is moved toward the extraction condition.

(47) In accordance with another aspect of the disclosed subject matter, and as further embodied herein, the pressure surface **1135** of the top member **1130** can include a roll-initiating notch **1155**. The notch **1155** can help direct the web material upwardly to initiate the formation of a roll with a small cross-sectional diameter within the feed space **1151**.

(48) In accordance with another aspect of the disclosed subject matter, and as embodied in the exemplary embodiment of FIG. **10A** and FIG. **10B**, the top member **1230** can be slidably connected to the base member **1210** and the first intermediate member **1220**. With reference to FIG. **10A**, the iris **1250** is configured to receive web material along the feed surface **1211** in a first condition. With reference to FIG. **10B**, the iris **500** is depicted in a second condition with the pressure surface **1232** of the top member **1230** moving away from the feed surface **1211**. In accordance with the disclosed subject matter, the cross dimension in side view between the feed surface **1211** and the pressure surface **1232** in the second condition is greater than the cross dimension in side view when the iris **1250** is in the first condition.

(49) For purpose of illustration, and not limitation, reference is made to the exemplary embodiment of an assembly **1500** shown in FIG. **11A** and FIG. **11B**. In accordance with another aspect of the disclosed subject matter, assembly **1500** includes a base member **1510**, a first intermediate member **1520** moveably coupled to the base member **1510**, and a top member **1530** moveably coupled to at least one of the base member **1510** or the first intermediate member **1520**. As embodied herein, the top member **1530** can be pivotally connected to the base member **1510**. The base member **1510** includes a concave arcuate feed surface **1505** defining an upwardly-extending ramp, and the top member **1530** includes a concave arcuate pressure surface **1535** facing the arcuate feed surface **1505**. The base member **1510**, first intermediate member **1520**, and the top member **1530**

collectively form an iris **1550** with a feed space **1551** defined between the feed surface **1505** and the pressure surface **1535**. Iris **1550** is depicted in broken line in FIGS. **11A** and **11B** for purpose of illustration only.

(50) The iris **1550** is moveable between a first condition and a second condition. With reference to FIG. **11A**, the exemplary assembly **1500** is depicted with the iris **1550** in a first condition. The feed space **1551** has a first cross dimension in side view between the feed surface **1505** and the pressure surface **1535** in the first condition. With reference to FIG. **11B**, the assembly **1500** is depicted in a second condition. The feed space **1551** has a second cross dimension in side view between the feed surface **1505** and the pressure surface **1535** in the second condition, the second cross dimension being greater than the first cross dimension. For purpose of example and as embodied herein, the second cross dimension can be approximately 4.5 inches. The iris **1550** in the first condition is configured to receive web material along the feed surface **1505** and direct the web material upwardly toward the top member **1535** to form a roll within the feed space. The iris **1550** is configured to move toward the second condition as the roll of web material increases in cross dimension against the pressure surface **1535** of the top member **1530**.

(51) As embodied herein, the first intermediate member **1520** can include an arcuate first intermediate pressure surface configured to further define the iris **1550** and apply pressure on a roll of web material as the roll increases in cross dimension and the iris **1550** expands from the first condition to the second condition. Additionally, and as further embodied herein, the assembly **1500** can include a second intermediate member **1540** moveably coupled to the base member **1510**. The second intermediate member **1540** can include an arcuate second intermediate pressure surface configured to further define the iris **1550** and apply pressure on a roll of web material as the roll increases in cross dimension and the iris **1550** expands from the first condition to the second condition.

(52) As embodied herein, the first intermediate member **1520**, the second intermediate member **1540**, and the top member **1530** can each be pivotally connected to the base member **1510**. As described above, the base member can include tiers or levels, and the various members can each be connected to a different tier of the base member to facilitate relative movement of the members. Additionally or alternatively, washers or spacers can be used at the connection between the base member **1510** and first intermediate member **1520** and between the base member **1510** and the second intermediate member **1540**, respectively, to offset the first intermediate member **1520**, second intermediate member **1540**, and top member **1530** from one another in end view and facilitate movement of the members. In accordance with another aspect of the disclosed subject matter, and as embodied herein, the first intermediate member **1520** and the second intermediate member **1540** can each include a region of increased thickness where the first intermediate member **1520** and the second intermediate member **1540**, respectively, connect to the base member **1510**. The regions of increased thickness can create an offset between the respective members to facilitate movement of the members.

(53) As described above, the top member **1530** and second intermediate member **1540** can each be slidably connected to the first intermediate member **1520**. The slidable connections between the top member **1530**, second intermediate member **1540**, and first intermediate member **1520** can further define the motion of the iris **1550** as the iris moves from the first condition to the second condition. As embodied herein, the top member **1530** can include a fastener **1534**, such as a bolt, screw, peg, pin, nail, or rivet, and the fastener **1534** can interface with a slot **1526** defined in the first intermediate member **1520** to slidably connect the top member **1530** to the first intermediate member **1520**. As further embodied herein, the first intermediate member **1520** can include a fastener **1524**, such as a bolt, screw, peg, pin, nail, or rivet, and the fastener **1524** can interface with a slot **1546** defined in the second intermediate member **1540** to slidably connect the second intermediate member **1540** to the first intermediate member **1520**.

(54) In accordance with an aspect of the disclosed subject matter, the top member **1530** of the

exemplary rolling assembly **1500** can include a ballast **1560**. For purpose of example, the top member **1530** can have a first end moveably coupled to at least one of the base member **1510** or the first intermediate member **1520** and a free end opposite the first end. The free end can include the ballast. As embodied herein, the first end of the top member **1530** can be pivotally connected to the base member **1510**. The ballast, or weight, **1560** can be selected to adjust the amount of pressure the iris applies to the roll as the iris moves from the first condition to the second condition.

Increasing the pressure the iris applies to a roll can, for example, help create tighter rolls of web material. The location and size of the ballast **1560** can be selected based on the desired performance characteristics of the assembly and the properties of the web material being rolled.

(55) In accordance with another aspect of the disclosed subject matter, the exemplary rolling assembly **1500** can include an outer member **1570** extending from the base member **1510**. For purpose of example and not limitation, outer member **1570** can be connected to the base member **1510** at an upper portion thereof. Outer member **1570** can include an arcuate outer member surface **1576**, which can further apply pressure to the roll of web material as the roll of web material increases in cross dimension within the iris. For purpose of example and as embodied herein, outer member **1570** can be connected to the base member **1510** using a threaded connector **1575**.

Connector **1575** can be used to adjust the position of the outer member **1575** relative to the base member **1510**. For purpose of example, the position of the outer member **1575** can be adjusted to define the desired diameter of a finished roll of web material. Additionally alternatively, the connector **1575** can be used to move the outer member **1575** away from the base member **1510** to remove a finished roll of web material from the iris **1550**. It is to be understood that the outer member **1575** is optional, and the rolling assembly **1500** can be used without the outer member **1575**.

(56) For purpose of illustration, and not limitation, reference is made to the exemplary embodiment of an assembly **1600** shown in FIG. **12A** and FIG. **12B**. In accordance with another aspect of the disclosed subject matter, assembly **1600** includes a base member **1610**, a first intermediate member **1620** moveably coupled to the base member **1610**, and a top member **1630** moveably coupled to at least one of the base member **1610** or the first intermediate member **1620**. As embodied herein, the top member **1630** can be pivotally connected to the base member **1610**. The base member **1610** includes a concave arcuate feed surface **1605** defining an upwardly-extending ramp, and the top member **1630** includes a concave arcuate pressure surface **1635** facing the arcuate feed surface **1605**. The base member **1610**, first intermediate member **1620**, and the top member **1630** collectively form an iris **1650** with a feed space **1651** defined between the feed surface **1605** and the pressure surface **1635**. Iris **1650** is depicted in broken line in FIGS. **12A** and **12B** for purpose of illustration only.

(57) The iris **1650** is moveable between a first condition and a second condition. With reference to FIG. **12A**, the exemplary assembly **1600** is depicted with the iris **1650** in the first condition. The feed space **1651** has a first cross dimension in side view between the feed surface **1605** and the pressure surface **1635** in the first condition. With reference to FIG. **12B**, the assembly **1600** is depicted in a second condition. The feed space **1651** has a second cross dimension in side view between the feed surface **1605** and the pressure surface **1635** in the second condition, the second cross dimension being greater than the first cross dimension. For purpose of example and as embodied herein, the second cross dimension can be approximately 2.5 inches. In accordance with another aspect of the disclosed subject matter, and as embodied herein, the assembly **1600** can include one or more features to adjust the size of the second cross dimension. For example, and as embodied herein, a stop **1612** can be included on the base member **1610**, and the stop can limit the movement of the iris when the desired cross dimension is reached.

(58) The iris **1650** in the first condition is configured to receive web material along the feed surface **1605** and direct the web material upwardly toward the top member **1635** to form a roll within the feed space. The iris **1650** is configured to move toward the second condition as the roll of web

material increases in cross dimension against the pressure surface **1635** of the top member.

(59) As embodied herein, the first intermediate member **1620** can include an arcuate first intermediate pressure surface configured to further define the iris **1650** and apply pressure on a roll of web material as the roll increases in cross dimension and the iris **1650** expands from the first condition to the second condition. Additionally, and as further embodied herein, the assembly **1600** can include a second intermediate member **1640** moveably coupled to the base member **1610**. The second intermediate member **1640** can include an arcuate second intermediate pressure surface configured to further define the iris **1650** and apply pressure on a roll of web material as the roll increases in cross dimension and the iris **1650** expands from the first condition to the second condition.

(60) As embodied herein, the first intermediate member **1620**, the second intermediate member **1640**, and the top member **1630** can each be pivotally connected to the base member **1610**. Additionally or alternatively, the top member **1630** and second intermediate member **1640** can each be slidably connected to the first intermediate member **1620**. The slidable connections between the top member **1630**, second intermediate member **1640**, and first intermediate member **1620** can further define the motion of the iris **1650** as the iris moves from the first condition to the second condition. As embodied herein, the top member **1630** can include a fastener **1631**, such as a bolt, screw, peg, pin, nail, or rivet, and the fastener **1631** can interface with a slot **1621** defined in the first intermediate member **1620** and a slot **1641** defined in the second intermediate member **1640** to slidably connect the top member **1630** and the second intermediate member **1640** to the first intermediate member **1620**.

(61) In accordance with another aspect of the disclosed subject matter, the top member **1630** of the exemplary rolling assembly **1600** can include a ballast **1660**. For purpose of example, the top member **1630** can have a first end moveably coupled to at least one of the base member **1610** or the first intermediate member **1620** and a free end opposite the first end. The free end can include the ballast. As embodied herein, the first end of the top member **1630** can be pivotally connected to the base member **1610**. As described above, the location and size of the ballast **1660** can be selected based on the desired performance characteristics of the assembly and the properties of the web material being rolled.

(62) In accordance with the disclosed subject matter, the base member **1610**, intermediate members **1620** and **1640**, and top member **1630** can be made of any suitable material using any suitable method of manufacture. For purpose of example, and not limitation, the members can be made of metal, such as steel or aluminum, polycarbonate, composites, such as cast polyurethane or plastic sheet materials such as Lexan and UEMWPE, or any other suitable material. In accordance with another aspect of the disclosed subject matter, and as embodied herein, the base member **1610** can be made of steel and the top member **1630**, first intermediate member **1620**, and second intermediate member **1640**, can be molded from composite material.

(63) Furthermore, the members can be coated or partially coated with desired materials to enhance performance, such as by reducing friction or increasing durability. For example, materials like Teflon and the like can be applied to reduce friction. Teflon can be applied or partially applied to the feed surface **1605** and/or pressure surface **1635** to reduce friction between the respective surface and web material to be rolled. The material selected for the various members can be chosen based on the desired performance characteristics of the rolling assembly. For example, selecting heavier materials for the top member can increase the pressure applied to the roll as the roll increases in cross dimension and the iris **1650** moves from a first condition to a second condition. Additionally, the material may be selected to achieve a desirable coefficient of friction between the members and the web material. For example the material of the members can be selected to prevent creation of scuffs or marks on the web as the material is rolled. In accordance with one aspect of the disclosed subject matter, the base member, intermediate members, and top member can be made of different materials from one another to achieve the desired performance characteristics.

(64) In accordance with the disclosed subject matter, the rolling assemblies for packaging web material in a roll described herein can be configured to receive web material **20** from a wide variety of delivery sources. As embodied herein, and with reference to the exemplary embodiment depicted in FIG. **8A**-FIG. **8C**, the iris **500** can be configured to receive a stream of web material from a feed assembly, such as a delivery belt or the like. For example, the feed assembly can be a vacuum belt, and a negative pressure can be used to keep the web material **20** secured to the belt as the belt transports the web material **20**. As embodied herein, as the iris **500** receives the web material **20** from the vacuum belt and directs the web material **20** from the first end **101** upwardly towards the top member **300**, the web material **20** can be released from the vacuum belt. The characteristics of the vacuum belt, such as the amount of negative pressure applied and the speed with which the vacuum belt delivers web material **20** to the iris **500**, can be selected as desired for the web material to be delivered. For purpose of example and as embodied herein, approximately 60 inches H₂O of negative pressure can be applied to the web material.

(65) The rolling assemblies for packaging web material in a roll can be used with a wide variety, types, and sizes of web material in accordance with the disclosed subject matter. For purpose of example, and not limitation, the web material can include polymeric packages, such as plastic bags or the like, or other stock materials, such as paper or cloth. Furthermore, the web material can include any suitable material, including paper, plastics, or composites. For example, the material can include polyethylene or polypropylene. As embodied herein, the web material of the disclosed subject matter can include plastic bags having a resealable closure mechanism, such as described in U.S. Pat. No. 6,450,686, the contents of which are hereby incorporated by reference in their entirety. For purpose of illustration, the web material, such as plastic bags, can be delivered to the iris **500** of the rolling assembly in a shingled arrangement. With reference to FIG. **8A**, the leading edge **21** of the web material **20** can include a first plastic bag, and a second plastic bag **23** can be positioned on top of the first plastic bag in a partially overlapping configuration such that a leading edge of the first bag is exposed. A third bag can be positioned on top of the second bag, and so on to create a shingled arrangement of bags.

(66) In accordance with another aspect of the disclosed subject matter, a system for packaging web material in a roll is provided. The system includes a feed assembly configured to carry a stream of web material and a rolling assembly proximate the feed assembly and configured to receive the stream of web material from the feed assembly. In accordance with the disclosed subject matter, the rolling assembly used with the system can be configured in accordance with any of the embodiments as described above or variations thereof. The rolling assembly includes a base member with a concave arcuate feed surface defining an upwardly-extending ramp and a first intermediate member moveably coupled to the base member. The assembly further includes a top member moveably coupled to at least one of the base member or the first intermediate member. The top member has a concave arcuate pressure surface facing the feed surface of the base member. The base member, the first intermediate member and the top member collectively form an iris with a feed space defined between the feed surface and the pressure surface. The iris is moveable between a first condition and a second condition, and the feed space has a first cross dimension in side view between the feed surface and the pressure surface in the first condition. The feed space has a second cross dimension in side view between the feed surface and the pressure surface when the iris is in the second condition. The second cross dimension is greater than the first cross dimension. The iris is configured in the first condition to receive web material along the feed surface and direct the web material upwardly toward the top member to form a roll within the feed space. The iris moves toward the second condition as the roll of web material increases in cross dimension against the pressure surface of the top member.

(67) In accordance with another aspect of the disclosed subject matter, and with reference to the exemplary embodiment of FIG. **13**, the feed assembly can include two vacuum belts **1301** and **1302** with a space therebetween, and the rolling assembly **1600** can be disposed in the space. A stream of

web material, such as a plurality of plastic bags in a shingled arrangement, can be carried on the vacuum belts **1301** and **1302** to the rolling assembly **1600**. As embodied herein, the feed surface **1605** of the assembly **1600** can be positioned to receive the stream of web material from the vacuum belts **1301** and **1302**. Additionally or alternatively, the feed assembly can be integrated with the base member **1610**.

(68) In accordance with another aspect of the disclosed subject matter, the rolling assembly **1600** can be mounted to a rotatable carriage assembly. For example, multiple rolling assemblies can be mounted to the carriage assembly. When a roll of web material is completed in the rolling assembly **1600**, the carriage assembly can rotate the rolling assembly **1600** away from the feed assembly to a position for extracting the completed roll from the rolling assembly. As the rolling assembly **1600** is rotated away from the feed assembly, the carriage assembly can rotate a second rolling assembly into place to receive a stream of web material from the feed assembly.

(69) The rolling assembly, system, and method can be used with a variety of sizes of web material. For example, and with reference to the exemplary embodiment of FIG. **14**, a plurality of rolling assemblies can be aligned in parallel to package wider webs in a roll. As embodied herein, the system for packaging web material in a roll can include a first rolling assembly **1402** and a second rolling assembly **1403** disposed proximate the feed assembly **1401** and configured to receive a stream of web material. The second rolling assembly can be used in tandem with the first rolling assembly to efficiently package web material in a roll depending on the properties of the web material to be packaged. For example, wide sheets of web material may benefit from the use of more than one rolling assembly.

(70) As embodied herein, the second rolling assembly **1403** can have the same configuration as the first rolling assembly **1402**. Alternatively, and in accordance with another aspect of the disclosed subject matter, the second rolling assembly **1403** can have a different configuration from the first rolling assembly **1402**. For example, rolling assemblies **1402** and **1403** having different iris cross-dimensions can be configured side by side to accommodate rollups having different diameters along the length of the roll. For example rolls of slider bags can include a larger diameter at the portion of the roll having the bag sliders and a smaller diameter at the portion of the roll without sliders. While the above description refers to the use of one or two rolling assemblies, any suitable number of rolling assemblies can be used.

(71) In accordance with another aspect of the disclosed subject matter, a method of packaging web material in a roll is provided. The method includes providing a rolling assembly configured to receive a stream of web material. In accordance with the disclosed subject matter, the rolling assembly provided can be configured in accordance with any of the embodiments as described above or variations thereof. The rolling assembly includes a base member with a concave arcuate feed surface defining an upwardly-extending ramp and a first intermediate member moveably coupled to the base member. The assembly further includes a top member moveably coupled to at least one of the base member or the first intermediate member. The top member has a concave arcuate pressure surface facing the feed surface of the base member. The base member, the first intermediate member and the top member collectively form an iris with a feed space defined between the feed surface and the pressure surface. The iris is moveable between a first condition and a second condition, and the feed space has a first cross dimension in side view between the feed surface and the pressure surface in the first condition. The feed space has a second cross dimension in side view between the feed surface and the pressure surface when the iris is in the second condition. The second cross dimension is greater than the first cross dimension.

(72) Methods in accordance with the disclosed subject matter further include delivering a stream of web material to the assembly. The iris receives the web material along the feed surface and directs the web material upwardly toward the top member to form a roll within the feed space. The iris moves toward the second condition as the roll of web material increases in cross dimension against the pressure surface of the top member.

(73) In accordance with another aspect of the disclosed subject matter, delivering the stream of web material can include delivering a plurality of plastic bags in a shingled arrangement.

(74) The assemblies, systems, and methods of the disclosed subject matter have demonstrated desirable performance characteristics not achieved with conventional devices and techniques. For example, currently known rolling assemblies are designed to maintain the final outside diameter of the roll of web material, but known assemblies do not maintain pressure on the roll of web material as the web material expands in cross dimension throughout the rolling process. As such, rolls of web material produced using known assemblies and methods can be less tightly rolled and have a larger final cross dimension for a given amount of web material than rolls produced according to the disclosed subject matter. Less tightly rolled web material can be less desirable, as the rolls with larger final cross dimension can require additional space for storage, transport, and commercialization. By contrast, rolling assemblies in accordance with the disclosed subject matter include an iris that is moveable between a first condition and a second condition, and a feed space defined between the feed surface of the base member and the pressure surface of the top member. As the roll of web material increases in cross dimension against the pressure surface of the top member, the iris maintains pressure on the roll of web material as the iris moves from the first condition to the second condition. As such, the rolling assemblies, systems, and methods in accordance with the disclosed subject matter can produce tighter rolls of web material than previously known rolling assemblies.

(75) Although the rolling assemblies herein are depicted in an orientation with the top member disposed above the base member such that gravity acts on the top member to bias the pressure surface of the top member towards the feed surface of the base member, it is to be understood that alternative configurations are envisioned within the scope of the disclosed subject matter. For example and not limitation, the pressure surface can be biased towards the feed surface with a spring, air cylinder, electric motor, or other mechanical or magnetic means.

(76) In addition to the specific embodiments claimed below, the disclosed subject matter is also directed to other embodiments having any other possible combination of the dependent features claimed below and those disclosed above. As such, the particular features presented in the dependent claims and disclosed above can be combined with each other in other manners within the scope of the disclosed subject matter such that the disclosed subject matter should be recognized as also specifically directed to other embodiments having any other possible combinations. Thus, the foregoing description of specific embodiments of the disclosed subject matter has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosed subject matter to those embodiments disclosed.

(77) It will be apparent to those skilled in the art that various modifications and variations can be made in the assemblies, systems, and methods of the disclosed subject matter without departing from the spirit or scope of the disclosed subject matter. Thus, it is intended that the disclosed subject matter include modifications and variations that are within the scope of the appended claims and their equivalents.

Claims

1. A rolling assembly for packaging web material in a roll, comprising: a base member having a concave arcuate feed surface defining an upwardly-extending ramp; a first intermediate member movably coupled to the base member; and a top member moveably coupled to at least one of the base member or the first intermediate member, the top member having a concave arcuate pressure surface facing the feed surface of the base member; wherein the base member, the first intermediate member and the top member collectively form an iris with a feed space defined between the feed surface and the pressure surface, the iris being moveable between a first condition and a second condition, the feed space having a first cross dimension in side view between the feed surface and

the pressure surface in the first condition and a second cross dimension in side view between the feed surface and the pressure surface in the second condition, the second cross dimension being greater than the first cross dimension, and the iris being configured in the first condition to receive web material along the feed surface and direct the web material upwardly toward the top member to form a roll within the feed space, the iris configured to move toward the second condition as the roll of web material increases in cross dimension against the pressure surface of the top member.

2. The rolling assembly of claim 1, wherein the first intermediate member is pivotally connected to the base member.
3. The rolling assembly of claim 1, wherein the rolling assembly further comprises a second intermediate member moveably coupled to the base member, the second intermediate member having an arcuate second intermediate pressure surface configured to further define the iris and apply pressure on the roll as the roll increases in cross dimension and the iris expands from the first condition to the second condition.
4. The rolling assembly of claim 3, wherein the first intermediate member, the second intermediate member, and the top member are each pivotally connected to the base member.
5. The rolling assembly of claim 3, wherein the top member and second intermediate member are each slidably connected to the first intermediate member.
6. The rolling assembly of claim 5, wherein the top member includes a first fastener and the first intermediate member includes a first slot, the first fastener interfacing with the first slot to slidably connect the top member and the first intermediate member.
7. The rolling assembly of claim 1, wherein the base member includes a stop, the stop configured to prevent further movement of the pressure surface away from the feed surface when the iris is in the second condition.
8. The rolling assembly of claim 1, wherein the top member includes a first end moveably coupled to at least one of the base member or the first intermediate member and a free end opposite the first end, the free end having a ballast.
9. The rolling assembly of claim 1, wherein the iris is configured to receive a stream of web material from a vacuum belt.
10. The rolling assembly of claim 1, wherein the web material includes a plurality of plastic bags in a shingled arrangement.
11. The rolling assembly of claim 1, wherein the material of the top member is selected from the group consisting of steel, aluminum, urethane or polycarbonate.
12. The rolling assembly of claim 1, wherein the pressure surface comprises Teflon.
13. A system for packaging web material in a roll comprising: a feed assembly configured to carry a stream of web material; a rolling assembly proximate the feed assembly and configured to receive a stream of web material from the feed assembly, the rolling assembly comprising: a base member having a concave arcuate feed surface defining an upwardly-extending ramp; a first intermediate member moveably coupled to the base member; and a top member moveably coupled to at least one of the base member and the first intermediate member, the top member having a concave arcuate pressure surface facing the feed surface of the base member; wherein the base member, the first intermediate member and the top member collectively form an iris with a feed space defined between the feed surface and the pressure surface, the iris being moveable between a first condition and a second condition, the feed space having a first cross dimension in side view between the feed surface and the pressure surface in the first condition and a second cross dimension between the feed surface and the pressure surface in the second condition, the second cross dimension being greater than the first cross dimension, and the iris being configured in the first condition to receive web material along the feed surface from the feed assembly and direct the web material upwardly toward the top member to form a roll within the feed space, the iris moving toward the second condition as the roll of web material increases in cross dimension against the pressure surface of the top member.

14. The system of claim 13, wherein the feed assembly comprises a vacuum belt.
 15. The system of claim 13, wherein the feed assembly comprises two vacuum belts with a space therebetween, and wherein the rolling assembly is disposed in the space.
 16. The system of claim 13, further comprising a second rolling assembly proximate the feed assembly and configured to receive the stream of web material.
 17. The system of claim 15 wherein the stream of web material includes a plurality of plastic bags in a shingled arrangement.
 18. A method for packaging web material in a roll, the method comprising: providing a rolling assembly configured to receive a stream of web material, the rolling assembly comprising: a base member having a concave arcuate feed surface defining an upwardly-extending ramp; a first intermediate member moveably coupled to the base member; and a top member moveably coupled to at least one of the base member and the first intermediate member, the top member having a concave arcuate pressure surface facing the feed surface of the base member; wherein the base member, the first intermediate member and the top member collectively form an iris with a feed space defined between the feed surface and the pressure surface, the iris being moveable between a first condition and a second condition, the feed space having a first cross dimension in side view between the feed surface and the pressure surface in the first condition and a second cross dimension between the feed surface and the pressure surface in the second condition, the second cross dimension being greater than the first cross dimension; and delivering a stream of web material to the assembly, the iris receiving the web material along the feed surface and directing the web material upwardly toward the top member to form a roll within the feed space, the iris moving toward the second condition as the roll of web material increases in cross dimension against the pressure surface of the top member.
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